VineWeave Canopy

green building | Group 12 | Integrated one-page design brief

Purpose and application

Provide a local shaded pause point in an open campus, pocket park or comparable public pedestrian space. The objective is to reduce direct solar exposure while keeping circulation, root space and maintenance access clear; this is not a claim of street-wide air cooling.

Geometry and parametric design

The tree-like, rounded shallow-arched support has 59 members, an overall extent of approximately 6.22 × 6.66 m and a maximum height of 3.48 m. The approximate maximum horizontal exterior span is 6.91 m. Ground-reaching support endpoint chords incline 45.9°–50.7° above horizontal; this is a defined geometric descriptor, not a local member angle.

Materials, planting and construction

Twelve galvanised Q355B Ø80×4 mm support members are proposed with 47 solid Ø80 mm engineered-bamboo crown members. Product identity and properties require confirmation. The source planting proposal uses two Boston ivy plants and one Rangoon creeper per unit, with local suitability, attachment and winter behaviour to be checked. Prefabricated numbered members connect through proposed head nodes, sleeves and clamps. Grouting requires curing; complete dry assembly and lossless reversibility are not established.

Environmental logic and analysis

The crown and foliage intercept direct sun; leaf placement and transmittance affect useful shade, while open sides retain potential airflow paths. Evapotranspiration depends on plant and water status. The 30°N example uses a 23.23 m^2 reference zone: ideal continuous-leaf solstice overlap is 47.2% / 98.7% / 47.2% at 09/12/15 solar time, with a 75.3% time mean. Seven parameter models and five spatial-cover states extend the analysis; these optical scenarios are neither plant-growth-duration predictions nor temperature reductions.

Implementation and evidence boundary

The current unit budget is approximately CNY 51,676 and the carbon estimate approximately 2,820 $kgCO_2e$, both preliminary. Final product factors, joint/global behaviour, wind and foundations, quotations and the actual site require checking. Structural acceptance and vegetation shade readiness are separate milestones; no evidence-based maturity month is yet available. The next step is to validate occupied-hour shade and environmental response under site-specific conditions.

Basis: project model and take-off (P1), ideal shade analysis (P2), Cost-Analysis.xlsx (P3), design/build-up (P4), parameter and spatial-cover supplement (P5).